MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIVIBRATOR, MONOSTABLE, DIGITAL, KEY CODE BME

This amendment forms a part of Military Specification MIL-M-28787/257(EC), dated 29 June 1982, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, 6, 7, 8, 9, AND 10

TABLE I: Delete and substitute new table I as printed on pages 2, 3, 4, 5, 6, 7, and 8 of this amendment.

PAGE 11

4.5.1, eighth line: Delete "100°C" and substitute "85°C".

6.4, second line: Delete "100°C" and substitute "85°C".

PAGE 12

6.4, Output drive current (V_{DD} = 5 V): Delete "100°C" (two places) and substitute "85°C" (two places).

PAGE 26

FIGURE 8, note 2: Delete "100°C" and substitute "85°C".

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FIGURE 9, note 2: Delete "100°C" and substitute "85°C".

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N521)

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FSC 5963

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B16	Input clamp diode voltage, negative	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C16	Input clamp diode voltage, positive	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D8	Output voltage, low level (I_{OL} = 0)	VI	X	0	50	0	50	0	50	0	50	0	50	mV
E1-E8	Output voltage, high level (I_{OH} = 0)	VII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
F1-F8	Output voltage, low level (V_{DD} = 5 V)	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
G1-G8	Output voltage, high level (V_{DD} = 5 V)	IX	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
H1-H8	Output voltage, low level (V_{DD} = 15 V)	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
J1-J8	Output voltage, high level (V_{DD} = 15 V)	XI	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
K1-K16	Input current, low level	XII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
L1-L16	Input current, high level	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
M1	Power supply current, low level	XIV	X	0	40	0	40	0	40	0	40	0	40	µA
M2	Power supply current, high level	XIV	X	0	40	0	40	0	40	0	40	0	40	µA
N1, N2	Output pulse width, match, V_{DD} = 5 V	XV	S	-400	400	-386	386	-416	416	-352	352	-426	426	ns

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See footnotes at end of table.

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Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N3, N4	Output pulse width, match, V_{DD} = 5 V	XV	S	-55	55	-53	53	-57	57	-48	48	-59	59	µs
P1, P2	Output pulse width, match, V_{DD} = 10 V	XVI	Q	-500	500	-482	482	-520	520	-440	440	-533	533	ns
P3, P4	Output pulse width, match, V_{DD} = 10 V	XVI	Q	-110	110	-106	106	-114	114	-97	97	-119	119	µs
Q1, Q2	Output pulse width, match, V_{DD} = 15 V	XVII	Q	-500	500	-482	482	-520	520	-440	440	-533	533	ns
Q3, Q4	Output pulse width, match, V_{DD} = 15 V	XVII	Q	-130	130	-126	126	-135	135	-114	114	-140	140	µs
R1-R4	Output pulse width check, V_{DD} = 5 V	XVIII	S	1.65	3.35	1.59	3.23	1.72	3.48	1.45	2.95	1.78	3.62	µs
R5-R8	Output pulse width check, V_{DD} = 5 V	XVIII	S	4.65	9.35	4.49	9.02	4.84	9.72	4.08	8.23	4.95	9.96	µs
R9-R12	Output pulse width check, V_{DD} = 5 V	XVIII	S	23.3	46.7	22.5	45.1	24.2	48.6	20.5	41.1	24.8	49.7	µs
R13-R16	Output pulse width check, V_{DD} = 5 V	XVIII	S	235	470	227	454	244	489	207	414	250	501	µs
S1-S4	Output pulse width check, V_{DD} = 10 V	XIX	Q	1.25	3.75	1.21	3.63	1.30	3.90	1.10	3.30	1.35	4.05	µs
S5-S8	Output pulse width check, V_{DD} = 10 V	XIX	Q	3.75	11.3	3.62	10.9	3.90	11.8	3.30	9.94	3.99	12.05	µs
S9-S12	Output pulse width check, V_{DD} = 10 V	XIX	Q	25	75	24	72	26	78	22	66	27	81	µs
S13-S16	Output pulse width check, V_{DD} = 10 V	XIX	Q	275	825	265	796	286	858	242	726	293	879	µs
T1-T4	Output pulse width check, V_{DD} = 15 V	XX	Q	1.25	3.75	1.21	3.62	1.30	3.90	1.10	3.30	1.35	4.05	µs

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T5-T8	Output pulse width check, V_{DD} = 15 V	XX	Q	4	12	3.9	11.6	4.2	12.5	3.5	10.6	4.3	12.8	µs
T9-T12	Output pulse width check, V_{DD} = 15 V	XX	Q	30	90	29	87	31	94	26	79	32	97	µs
T13-T16	Output pulse width check, V_{DD} = 15 V	XX	Q	325	975	314	941	338	1014	286	858	346	1038	µs
U1-U8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XXI	S	35	350	35	350	37	375	29	292	40	393	ns
U9-U16	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XXI	Q	83	830	83	830	90	906	65	657	99	960	ns
V1-V8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XXII	S	26	260	26	260	27	273	23	231	28	282	ns
V9-V16	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XXII	Q	74	740	74	740	77	778	65	654	82	805	ns
W1-W8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XXIII	Q	15	150	15	150	16	161	12	126	17	169	ns
W9-W16	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XXIII	Q	13	130	13	130	13	135	11	118	14	141	ns
X1-X8	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XXIV	Q	11	110	11	110	11	118	9	92	12	127	ns
X9-X16	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XXIV	Q	10	100	10	100	10	104	9	90	10	109	ns
Y1-Y8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XXV	S	66	660	66	660	67	671	63	636	68	638	ns

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						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y9-Y16	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XXV	S	110	1100	110	1100	111	1110	107	1076	112	1118	ns
Y17-Y24	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XXV	Q	86	860	86	860	89	892	78	788	92	915	ns
Y25-Y32	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XXV	Q	130	1300	130	1300	133	1335	122	1228	136	1356	ns
Z1-Z8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXVI	S	66	660	66	660	67	671	63	636	68	678	ns
Z9-Z16	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXVI	S	110	1100	110	1100	111	1110	107	1076	112	1118	ns
Z17-Z24	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXVI	Q	86	860	86	860	89	892	78	788	92	915	ns
Z25-Z32	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXVI	Q	130	1300	130	1300	133	1332	122	1228	136	1356	ns
AA1-AA8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXVII	Q	23	230	23	230	23	235	21	219	24	240	ns
AA9-AA16	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXVII	Q	45	450	45	450	45	455	43	439	46	460	ns
BB1-BB8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXVIII	Q	23	230	23	230	23	235	21	219	24	240	ns
BB9-BB16	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXVIII	Q	45	450	45	450	45	455	43	439	46	460	ns

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						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
CC1-CC8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXIX	Q	17	175	17	175	17	179	16	165	18	184	ns
CC9-CC16	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXIX	Q	32	325	32	325	32	329	31	315	34	338	ns
DD1-DD8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXX	Q	17	175	17	175	17	179	16	165	18	184	ns
DD9-DD16	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXX	Q	32	325	32	325	32	329	31	315	34	338	ns
EE1-EE4	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXXI	S	66	660	66	660	67	670	63	636	68	678	ns
EE5-EE8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXXI	S	110	1100	110	1100	111	1110	107	1076	112	1118	ns
EE9-EE12	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXXI	Q	86	860	86	860	89	892	78	788	92	915	ns
EE13-EE16	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXXI	Q	130	1300	130	1300	133	1332	122	1228	136	1356	ns
FF1-FF4	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XXXII	S	66	660	66	660	67	670	63	636	68	678	ns
FF5-FF8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XXXII	S	110	1100	110	1100	111	1110	107	1076	112	1118	ns
FF9-FF12	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XXXII	Q	86	860	86	860	89	892	78	788	92	915	ns

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						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
FF13-FF16	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XXXII	Q	130	1300	130	1300	133	1332	122	1228	136	1356	ns
GG1-GG4	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXXIII	Q	23	230	23	230	23	235	21	219	24	240	ns
GG5-GG8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXXIII	Q	60	600	60	600	60	605	58	589	61	610	ns
HH1-HH4	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXXIV	Q	23	230	23	230	23	235	21	219	24	240	ns
HH5-HH8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXXIV	Q	60	600	60	600	60	605	58	589	61	610	ns
JJ1-JJ4	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXXV	Q	17	175	17	175	17	179	16	165	18	184	ns
JJ5-JJ8	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXXV	Q	50	500	50	500	50	504	49	490	51	509	ns
KK1-KK4	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXXVI	Q	17	175	17	175	17	179	16	165	18	184	ns
KK5-KK8	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXXVI	Q	50	500	50	500	50	504	49	490	51	509	ns
LL1-LL4	Output retriggerability check, V_{DD} = 5 V	XXXVII	X	1.65	6.70	1.59	6.47	1.72	6.97	1.45	5.90	1.79	7.14	µs
MM1-MM4	Output retriggerability check, V_{DD} = 10 V	XXXVIII	S	1.25	7.50	1.21	7.24	1.30	7.80	1.10	6.60	1.35	7.99	µs

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						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
NN1-NN4	Output retriggerability check, V_{DD} = 15 V	XXXIX	S	1.25	7.50	1.21	7.24	1.30	7.80	1.10	6.60	1.35	7.99	µs
PP1-PP4	Resistance	XL	X	9.9	10.1	9.9	10.1	9.8	10.2	9.9	10.1	9.6	10.4	kΩ
QQ	Power supply filter capacitance	XLI	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
RR1-RR4	Capacitance	XLII	S	198	242	198	242	198	242	198	242	198	242	pF
SS1-SS16	Input capacitance	XLIII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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